Perlims -1
Marks 80
Time :− 3 hrs

ICSE : X
Date :

MATHS BRIDGE ACADEMY

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SECTION A

(Attempt all questions from this section.)

Question-1 Choose the correct answers to the questions from the given options. [15]

(i) The given quadratic equation $2x^2 - 9x + 13 = 0$ has

(a) Two equal roots
(b) Real & unequal roots
(c) Only one real root
(d) No real roots.

(ii) Sanjana opens a recurring deposit scheme & deposits ₹ 800 per month for a period of 2 yea₹. If the rate of interest is 9% p.a., & the interest earned is ₹ 1800, then the matured amount at the end of 2 yea₹ is

(a) ₹ $(24 \times 800) +$ ₹ 1800
(b) ₹ $(2 \times 800) +$ ₹ 1800
(c) ₹ $(4 \times 800) +$ ₹ 1800
(d) ₹ $(12 \times 800) +$ ₹ 1800

(iii) 20 tickets, on which numbers 1 to 20 are written & mixed thoroughly, then a ticket is drawn randomly out of them. What is the probability the number drawn is a multiple of 3 or 7?

(a) $\frac{1}{5}$ (b) $\frac{1}{4}$ (c) $\frac{2}{5}$ (d) 0

(iv) (A): The reflection of the point P$(-2, 7)$ in the y-axis is $(2, 7)$.
(R): When a point is reflected in y-axis, only the sign of y-coordinate changes.

(v) (A): Δ ABC ~ Δ DEF such that A(Δ ABC) $= 36\ cm^2$ & A(Δ DEF) $= 49\ cm^2$ then AB : DE = 6 : 7.
(R): If Δ ABC ~ Δ DEF, then $\frac{A(ABC)}{A(DEF)} = \frac{AB^2}{DE^2} = \frac{BC^2}{EF^2} = \frac{AC^2}{DF^2}$.

(a) Assertion (A) is true but reason (R) is false.
(b) Assertion (A) is false but reason (R) is true.
(c) Both assertion (A) & reason (R) are true & reason (R) is the correct explanation of assertion (A).
(d) Both assertion (A) & reason (R) are true & reason (R) is not the correct explanation of (A).

(vi) If $A = \begin{bmatrix} 3 & 9 \\ 7 & 2 \end{bmatrix}$ & I is an identity matrix of order 2, then the value of A + 5I is

(a) $\begin{bmatrix} 3 & 9 \\ 7 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} 15 & 45 \\ 35 & 10 \end{bmatrix}$ (c) $\begin{bmatrix} 8 & 14 \\ 12 & 7 \end{bmatrix}$ (d) $\begin{bmatrix} 8 & 9 \\ 7 & 7 \end{bmatrix}$

(vii) QP is a tangent to a circle with centre O at a point P on the circle. If Δ OPQ is isosceles, then angle OQP equals to

(a) 30 ° (b) 45 ° (c) 60 ° (d) 90 °

(viii) If both $(x - 2)$ & $\left(x - \frac{1}{2}\right)$ are the facto₹ of $px^2 + 5x + r$, then:

(a) p = 2r (b) 2p = r (c) p = r (d) p = r + 2

(ix) The money required to buy 100, ₹ 50 shares quoted at ₹ 48.50 is:

(a) ₹ 5000 (b) ₹ 2425 (c) ₹ 2500 (d) ₹ 4850

(x) The common difference of the AP $\frac{1}{p}, \frac{1-p}{p}, \frac{1-2p}{p}$ is

(a) 1 (b) $\frac{1}{p}$ (c) -1 (d) $-\frac{1}{p}$

(xi) If $\sin\theta + \cos\theta = \sqrt{2}\cos\theta$ then the value of tan(theta) is

(a) $\sqrt{2}+1$ (b) $\sqrt{2}-1$ (c) $\sqrt{2}$ (d) $-\sqrt{2}$

(xii) A sphere is melted & half of the melted liquid is used to form 11 identical cubes, whereas the remaining half is used to form 7 identical smaller spheres. The ratio of the side of the cube to the radius of the new small sphere is

(a) $\left(\frac{4}{3}\right)^{\frac{1}{3}}$ (b) $(4)^{\frac{1}{3}}$ (c) $\left(\frac{8}{3}\right)^{\frac{1}{3}}$ (d) $(3)^{\frac{1}{3}}$

(xiii) If $3x + 4y - k = 0$ is a median through A of the triangle whose vertices are at points A(2, 4), B(4, 3) & C(8, $-$ 1), then the value of k is

(a) -22 (b) 22 (c) 11 (d) -11

(xiv) Statement 1: The solution set of $7x + 3 < 5x + 9, x \in$ R is $\{x : x < 3, x \in R\}$

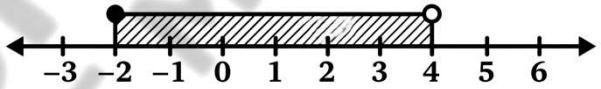
Statement 2: The graph of the solution of the above inequality is represented on the number line.

(a) Both the statements are true.

(b) Both the statements are false.

(c) Statement 1 is true & statement 2 is false.

(d) Statement 1 is false & statement 2 is true.



(xv) Which of the following cannot be determined graphically for a grouped frequency distribution?

(a) Mean (b) Median (c) Quartiles (d) Mode

Question-2

(i) Solve the quadratic equation $5x^2 - 3x - 4 = 0$, giving your answer correct to 3 significant figures. (Use mathematical tables for this question) [4]

(ii) Mohan has a recurring deposit account in a bank for 2 years at 6% p.a. simple interest. If he gets ₹ 1200 as interest at the time of maturity, find:

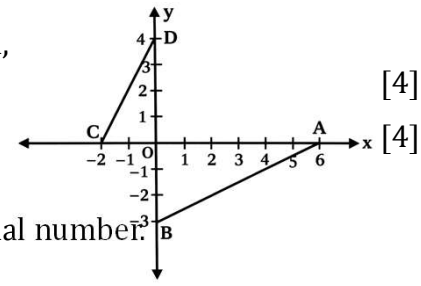
i. the monthly instalment. ii. the amount at the time of maturity. [4]

(iii) Find the ratio in which (4, b) divides the line joining the points (6, $-$ 2) & ($-$ 3, 16). Also find b. [4]

Question-3

(i) Construct a cyclic quadrilateral ABCD in which AB = 5 cm, BC = 8 cm, ∠ABC = 60 ° & D is equidistant from B & C. [4]

(ii) From the given graph, prove that Δ AOB ~ Δ DOC. Find $\frac{\text{A}(\Delta\,\text{AOB})}{\text{A}(\Delta\,\text{DOC})}$. [4]



(iii) The digits of a 3-digit positive number are in AP & their sum is 15. The number obtained by reversing its digits is 792 more than the original number. Find the original number.

SECTION B (Attempt any four questions from this section.)

Question-4

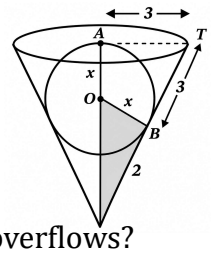
(i) Find the solution set of the following inequation and represent it on the number line, where x in I:

$-2\left(\frac{1}{6}\right) \le \left(\frac{x}{3}\right) - 1\left(\frac{1}{6}\right) < \frac{5}{6}$ [3]

(ii) Given $\begin{bmatrix} 2 & 1 \\ -3 & 4 \end{bmatrix} . X = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$ i. Write the order of matrix X. ii. Find the matrix X. [3]

(iii) Prove that: $\frac{\cot^2 A}{(cosec(A)+1)^2} = \frac{(1-\sin A)}{(1+\sin A)}$ [4]

Question-5



(i) Using properties of proportion, solve for x: $\frac{(x^2+x-3)}{(x-3)} = \frac{(4x^2+3x-2)}{(3x-2)}$ [3]

(ii) Using the Remainder theorem, factorise completely the following polynomial:
$3x^3 + 2x^2 - 19x + 6$ [3]

(iii) A cone of radius 3 cm and height 4 cm is full of water. When a sphere which exactly fits and touches the sides of the cone is immersed, what fraction of water overflows? [4]

Question-6

(i) Points (0, 4), (0, -2) are invariant on reflection in line L1 and points (1, 0), (-3, 0) are invariant on reflection in line L2.

i. Name the lines L1 and L2 and write their equations.

ii. Write the image of P(-3, 4) in L1. Name the image as P' and write its co-ordinates.

iii. Reflect P' in L2 and name it P'' and hence write its co-ordinates.

iv. State a single transformation that maps P onto P''. [5]

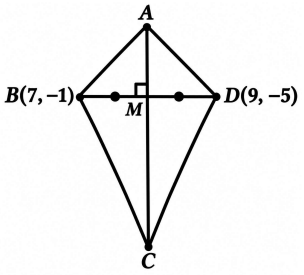
(ii) A boy is flying a kite with a string of length 80 m. The angle of elevation of the kite is found to be 30 °. On the other side of the kite, a girl standing near the window at a height of 16 m observes that the angle of elevation of the kite is 60 °. Find the distance between the girl and the kite. [5]

Question-7

(i) Find the mode of the following distribution by drawing a histogram. [3]

Wages in ₹:	40-50	50-60	60-70	70-80	80-90	90-100
No. of workers:	3	8	12	6	4	2

(ii) ABCD is a kite, B = (7, -1) and D = (9, -5) form the shorter diagonal. Find the equation of the longer diagonal AC. [3]



(iii) Pramod bought the following articles from the departmental store:

Sr. No.	Item	Price	Rate of GST
1	Article A	₹ 4000	18%
2	Article B	₹ 2400	5%

[4]

He gets a discount of 5% on the article costing more than ₹ 2000 and he paid ₹ 7000 cash to the cashier of the departmental store towards his bill. Find how much extra did Pramod pay to the cashier.

Question-8

(i) Priyanka invested ₹ 8000 in 7%, ₹ 100 shares at ₹ 80. After a year she sold these shares at ₹ 75 each and invested the proceeds (including her dividend) in 18%, ₹ 25 shares at ₹ 41. Find

i. her dividend for the first year. ii. her annual income on shares in the second year.

iii. the percentage increase in her return on her original investment.

(ii) Using step deviation method, calculate the mean marks of the following distribution.

Class-interval	50-55	55-60	60-65	65-70	70-75	75-80	80-85	85-90
Frequency:	5	20	10	10	9	6	12	8

(iii) Five years ago, a woman's age was the square of her son's age. Four years hence, her age will be thrice that of her son's age. Find i. the age of son five years ago. ii. the present age of the woman.

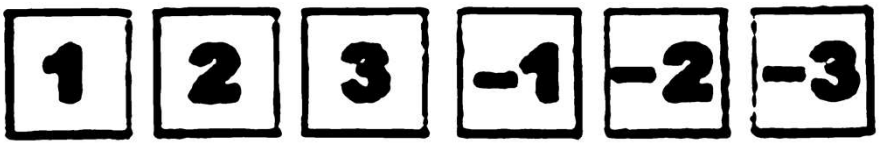
MATHS BRIDGE ACADEMY

Question-9

(i) Small 126 cones, each of diameter 7 cm and height 3 cm, are melted and recast into a metallic sphere. Find the diameter of the sphere and hence find its surface area. Use $\pi = \frac{22}{7}$. [3]

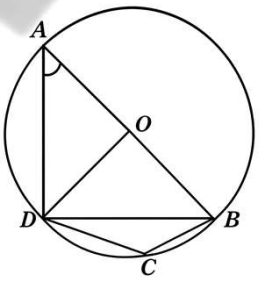
(ii) A die has 6 faces marked by given numbers

. The die is thrown once, what is the probability of getting [3]



i. A positive integer? ii. An integer greater than -3? iii. The smallest integer?

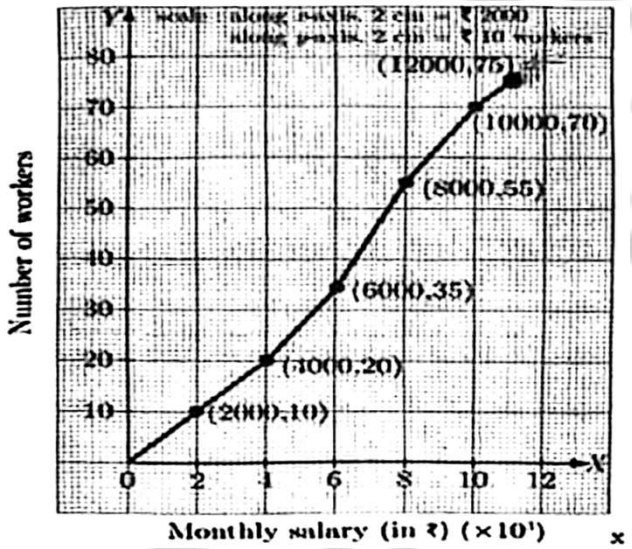
(iii) In the given figure ABCD is a cyclic quadrilateral with centre O.
Here OD || BC and ∠DAB = 55 °.
Find i.∠BCD. ii. ∠ BOD iii. ∠ ODB iv. ∠BDC. [4]



Question-10

(i) The fifth term of a GP is 48 and the eighth term is 384.
Find its 12th term. [3]

(ii) Using graph answer the following questions:



Scale: along x-axis, 2 cm = ₹ 2000 along y-axis, 2 cm = 10 workers
Find: i. Total number of workers. ii. The median class. [3]
iii. Number of workers having monthly salary more than or equal to ₹ 6,000 but less than ₹ 10,000.

(iii) In the circle with centre O, AT and BT are tangents. If ∠T = 70 °,
find ∠AOB, ∠APB, ∠AQB and ∠TAB. [4]

